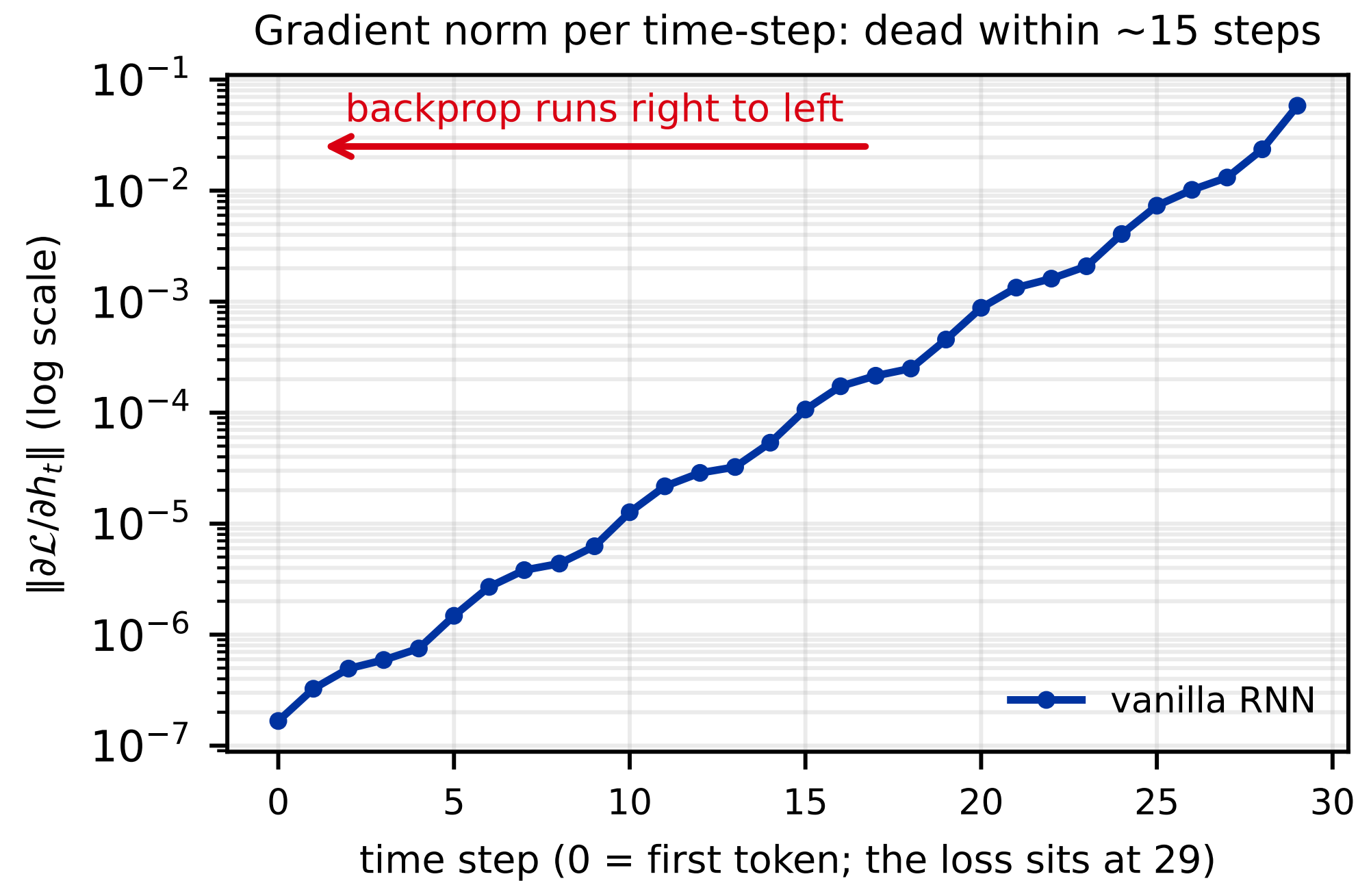
Gradient norm per time-step: dead within ~15 steps
backprop runs right to left
$\|\partial \mathcal{L}/\partial h_t\|$ (log scale)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
vanilla RNN
0
5
10
15
20
25
30
time step (0 = first token; the loss sits at 29)